

Part II V4 in Lamport notation

Fixed-time comparison, spectral transfer, ground-state limit, and assembly

Fresh reconstruction of `part_ii/v4_assembly.tex`

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This module proves the V4 assembly from the corrected V1–V3 modules. In particular, every use of V2’s shifted estimate refers to the repaired fundamental-band/alias proof, and every use of its fourth-chaos comparison includes the four-edge term omitted by the source V2. Thus the argument below validates the claimed endpoint after those repairs; it does not retroactively validate the original V2 proof.

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1 Objects and corrected input contract

Fix $L, m > 0$, $\lambda \geq 0$, a coarse scale M , and $\xi = \varepsilon_M q + ip \in h_{M,L}$. Let $N = M + r$, $\varepsilon_N = 2^{-r} \varepsilon_M$, and

$$\mathcal{L}_N = 1 + \log(1/\varepsilon_N), \quad \eta_D = 2 - \log_2 3 > 0. \quad (0.1)$$

The lattice Hamiltonian $K_{N,L}$ has ordered eigenvalues $E_0(N) < E_1(N) \leq \dots$, simple ground vector Ω_N , and ground state ω_N . Put

$$e_0(N) = \frac{1}{2} \sum_{k \in \Gamma_N} \gamma_N(k), \quad \tilde{E}_j(N) = E_j(N) - e_0(N), \quad (0.2)$$

$$D_N(t) = \sum_{j \geq 1} e^{-t(E_j(N) - E_0(N))}. \quad (0.3)$$

The continuum operator K_L constructed in the V3 module has eigenvalues $\mu_0 < \mu_1 \leq \dots$, normalized positive ground vector Ω_L , gap $\gamma_L = \mu_1 - \mu_0 > 0$, and

$$D_\infty(t) = \sum_{j \geq 1} e^{-t(\mu_j - \mu_0)} < \infty. \quad (0.4)$$

Define the thermal Weyl functionals

$$F_N(t; \xi) = \frac{\text{Tr}(e^{-tK_{N,L}} W_N(R_M^N \xi))}{\text{Tr} e^{-tK_{N,L}}}, \quad F_\infty(t; \xi) = \frac{\text{Tr}(e^{-tK_L} W_{\text{ct}}(R_M^\infty \xi))}{\text{Tr} e^{-tK_L}}. \quad (0.5)$$

Both have modulus at most one because the heat operator is positive trace class and the Weyl operator is unitary.

The input contract is as follows.

1. V1 supplies the exact lattice marked-slice mean-shift formula, with the free periodic Gaussian reference and with the interaction, not the full harmonic potential, in the path weight.
2. The repaired V2 module supplies the common-noise laws, unshifted comparison, the explicit four-edge correction, the exact fundamental-band/alias shifted comparison, and uniform plain/shifted Nelson bounds. Its standing time hypothesis is $t_0 L \geq 1$.
3. V3 supplies the continuum trace, marked-Weyl formula, trace-class semigroup, simple ground state, and positive gap.
4. The earlier free-rate and Weyl-reduction Lamport modules supply their stated elementary estimates and C^* -algebraic extension theorem.

No new printed theorem is imported in this module.

2 Exact coupled representations

Work on the repaired V2 common-noise probability space. Let η_N have the lattice periodic law and η the continuum periodic law. Let $\mathcal{U}_{N,t}, \mathcal{U}_t$ be the corresponding interactions, and $h^{(N)}, h^{(\infty)}$ the deterministic jump interpolants. Define

$$b_N(k) = \varepsilon_M^{1/2} P_r(\varepsilon_M k) \hat{p}_M(k), \quad a_N(k) = \varepsilon_M^{1/2} P_r(\varepsilon_M k) \hat{q}_M(k), \quad (1.1)$$

and define a_∞, b_∞ by replacing P_r with P_∞ .

$\langle 1 \rangle$ 1 Write both thermal functionals on this single coupling space.

$\langle 2 \rangle$ 1 Define

$$S_N = \frac{1}{8L} \sum_{k \in \Gamma_N} \gamma_N(k) \coth(t\gamma_N(k)/2) |b_N(k)|^2, \quad S_\infty = \frac{1}{8L} \sum_{k \in \Gamma_\infty} \gamma(k) \coth(t\gamma(k)/2) |b_\infty(k)|^2, \quad (1.2)$$

$$\Phi_N(J_q) = \varepsilon_N \sum_{x \in \Lambda_N} q_N(x) \eta_N(0, x), \quad \Phi(J_q) = \int_{\mathbb{T}_L} q_\infty(x) \eta(0, x) dx. \quad (1.3)$$

$\langle 2 \rangle$ 2 Put

$$\mathcal{N}_N = \mathbb{E}[e^{i\Phi_N(J_q)} e^{-\mathcal{U}_{N,t}(\eta_N + h^{(N)})}], \quad \mathcal{E}_N = \mathbb{E}[e^{-\mathcal{U}_{N,t}}], \quad (1.4)$$

$$\mathcal{N}_\infty = \mathbb{E}[e^{i\Phi(J_q)} e^{-\mathcal{U}_t(\eta + h^{(\infty)})}], \quad \mathcal{E}_\infty = \mathbb{E}[e^{-\mathcal{U}_t}]. \quad (1.5)$$

$\langle 2 \rangle$ 3 The exact representations are

$$F_N = e^{-S_N} \frac{\mathcal{N}_N}{\mathcal{E}_N}, \quad F_\infty = e^{-S_\infty} \frac{\mathcal{N}_\infty}{\mathcal{E}_\infty}. \quad (1.6)$$

Because: The first is V1’s mode-by-mode Mehler mean-shift identity with the corrected interaction reference. The second is equation (6.8) of the V3 module. The repaired V2 covariance proposition identifies their Gaussian fields on one space. The Appell formula in V1/V3 shows that no extra Wick-subtraction term is present. All factors and normalizations appear explicitly in (1.1)–(1.5).

⟨2⟩ 4 This proves ⟨1⟩1. □

3 Action and marked-source comparisons

Throughout this section, $t \in [t_0, T]$, $t_0 L \geq 1$, and constants may depend on the fixed coarse data but not on N or t . Put $\bar{w} = \coth(t_0 m/2)$.

⟨1⟩ 2 Compare the action prefactors.

⟨2⟩ 1 $0 \leq S_N, S_\infty \leq C_\xi$.

Because: The D4 envelope gives

$$|b_\bullet(k)|^2 \leq C\varepsilon_M \|\widehat{p}_M\|_\infty^2 (1 + \varepsilon_M |k|)^{-2-\eta_D}. \quad (2.1)$$

Also $\gamma_N \leq \gamma$, $\coth(t\gamma_N/2) \leq \bar{w}$, and $\sum_k (m + |k|)(1 + \varepsilon_M |k|)^{-2-\eta_D} < \infty$.

⟨2⟩ 2 Let $\ell = \varepsilon_M |k|$ and split at $\ell_r = 2^{r/2}$. On $\ell \leq \ell_r$,

$$|\gamma_N \coth(t\gamma_N/2) - \gamma \coth(t\gamma/2)| \leq C(t_0, m) \varepsilon_N^2 k^4, \quad (2.2)$$

$$|b_N|^2 - |b_\infty|^2 \leq C_\xi 2^{-r} \ell (1 + \ell)^{-2-\eta_D}. \quad (2.3)$$

Because: For $w(y) = y \coth(ty/2)$, $|w'(y)| \leq \bar{w} + \sup_{x \geq t_0 m/2} x / \sinh^2 x$. The lattice dispersion estimate gives $0 \leq \gamma - \gamma_N \leq \varepsilon_N^2 k^4 / (24m)$, proving (2.2). The QMF telescope gives $|P_r - P_\infty| \leq C 2^{-r} \ell (1 + \ell)^{-1-\eta_D/2}$; multiply by $|b_N| + |b_\infty|$ and use (2.1) to obtain (2.3).

⟨2⟩ 3 Summing the low-mode dispersion term gives

$$C 4^{-r} \int_0^{\ell_r} \ell^4 (1 + \ell)^{-2-\eta_D} d\ell \leq C 2^{-r(1+\eta_D)/2}. \quad (2.4)$$

Because: The mode density in ℓ is constant. At large ℓ , the integrand is $\ell^{2-\eta_D}$, whose integral is $O(\ell_r^{3-\eta_D})$; multiply by 4^{-r} .

⟨2⟩ 4 Summing the low-mode filter term gives

$$C 2^{-r} \int_0^{\ell_r} \ell^2 (1 + \ell)^{-2-\eta_D} d\ell \leq C 2^{-r(1+\eta_D)/2}. \quad (2.5)$$

Because: The action weight grows at most linearly in ℓ ; (2.3) supplies the other factor. The integral is $O(\ell_r^{1-\eta_D})$.

⟨2⟩ 5 The high-mode tails of both actions obey

$$C \int_{\ell_r}^\infty \ell (1 + \ell)^{-2-\eta_D} d\ell \leq C 2^{-r\eta_D/2}. \quad (2.6)$$

⟨2⟩ 6 Therefore

$$|S_N - S_\infty| \leq C_\xi 2^{-r\eta_D/2}, \quad |e^{-S_N} - e^{-S_\infty}| \leq |S_N - S_\infty|. \quad (2.7)$$

Because: Add (2.4)–(2.6). The last inequality follows from the mean-value theorem because $S_N, S_\infty \geq 0$ and $|(-e^{-x})| \leq 1$ there.

$\langle 2 \rangle 7$ This proves $\langle 1 \rangle 2$. □

$\langle 1 \rangle 3$ Compare the marked-slice source phases, including the asymmetric edge.

$\langle 2 \rangle 1$ For $r \geq 1$, $a_N(\pm\pi/\varepsilon_N) = 0$.

Because: The refinement mask contains the factor $m_0(\pm\pi) = 0$ at the last scale, so $P_r(\pm\pi 2^r) = 0$. Thus the symmetrized V2 edge noise makes no contribution to the linear marked source. This fact applies only to this linear source; it does not erase edge terms from covariance powers.

$\langle 2 \rangle 2$ For $r \geq 1$, orthogonality of the common Gaussian Fourier family gives

$$\mathbb{E}[(\Phi_N - \Phi)^2] = \frac{1}{2Lt} \left[\sum_{\kappa \in D_N} |c_N(\kappa)^{1/2} a_N(-k) - c_\infty(\kappa)^{1/2} a_\infty(-k)|^2 + \sum_{\kappa \in D_\infty \setminus D_N} c_\infty(\kappa) |a_\infty(-k)|^2 \right]. \quad (2.8)$$

Because: After Step $\langle 2 \rangle 1$, the lattice source contains no edge coefficient, so its Fourier noise is literally the restriction of the common family. Subtract the two series and apply the Gaussian isometry. The edge mode $+\pi/\varepsilon_N$ is included in the continuum-tail sum.

$\langle 2 \rangle 3$ The filter-difference part of the first sum in (2.8) is at most

$$C_\xi 4^{-r} \int_0^{\pi 2^r} \ell^2 (1 + \ell)^{-2-\eta_D} d\ell \leq C_\xi 2^{-r(1+\eta_D)}. \quad (2.9)$$

Because: $t^{-1} \sum_{k_0} c_N(k_0, k) = \coth(t\gamma_N/2)/(2\gamma_N) \leq \bar{w}/(2m)$; square the QMF filter difference used in (2.3) and sum the spatial modes.

$\langle 2 \rangle 4$ The covariance-symbol part is $O(\varepsilon_N^2)$.

Because: $(\sqrt{x} - \sqrt{y})^2 \leq |x - y|$ and the repaired V2 symbol estimate is

$$|c_N - c_\infty|(k_0, k) \leq C\varepsilon_N^2 k^4 (m^2 + k_0^2 + k^2)^{-2}. \quad (2.10)$$

The exact k_0 count gives

$$t^{-1} \sum_{k_0} |c_N - c_\infty| \leq C\varepsilon_N^2 \frac{k^4}{(m^2 + k^2)^{3/2}} \leq C\varepsilon_N^2 (m + |k|).$$

Multiplying by the D4 envelope for $|a_\infty|^2$ leaves a summable $|k|^{-1-\eta_D}$ tail.

$\langle 2 \rangle 5$ The missing-mode term in (2.8) is $O(2^{-r(1+\eta_D)})$.

Because: Sum $t^{-1} \sum_{k_0} c_\infty \leq \bar{w}/(2\gamma(k))$ against $|a_\infty(k)|^2$ for $|k| \geq \pi/\varepsilon_N$ and use the D4 tail.

$\langle 2 \rangle 6$ After increasing the constant to cover the one fixed case $r = 0$,

$$\mathbb{E}[(\Phi_N - \Phi)^2] \leq C_\xi 2^{-r\eta_D/2}, \quad \|e^{i\Phi_N} - e^{i\Phi}\|_2 \leq C_\xi 2^{-r\eta_D/4}. \quad (2.11)$$

Because: For $r \geq 1$, combine (2.9)–(2.10) and the missing tail, then weaken the exponent to a common one. At $r = 0$ the replacement of the edge noise in (2.8) is not valid, but both variances are finite and bounded by a constant depending on the fixed coarse scale, which equals the right side after enlarging C_ξ . Finally use $|e^{ix} - e^{iy}| \leq |x - y|$.

$\langle 2 \rangle 7$ This proves $\langle 1 \rangle 3$. □

4 Numerators, denominators, and fixed-time convergence

⟨1⟩ 4 Bound the denominators away from zero.

⟨2⟩ 1 At the gate endpoint $(a, b) = (0, 0)$,

$$\mathbb{E}\mathcal{U}_{N,t} = 3\lambda d_N(t)^2(2Lt), \quad \mathbb{E}\mathcal{U}_t = 3\lambda d_\infty(t)^2(2Lt). \quad (3.1)$$

Because: The ensemble-Wick chaoses of orders two and four are centered. The Wick recombination constant term is exactly $3d^2$; integrate the constant over volume $2Lt$.

⟨2⟩ 2 Since $0 \leq d_N, d_\infty \leq \bar{d}$,

$$\mathcal{E}_N, \mathcal{E}_\infty \geq e^{-C_0}, \quad C_0 = 3\lambda \bar{d}^2(2LT). \quad (3.2)$$

Because: Jensen's inequality for the convex function e^{-x} gives $\mathbb{E}e^{-\mathcal{U}} \geq e^{-\mathbb{E}\mathcal{U}}$; insert (3.1).

⟨2⟩ 3 This proves ⟨1⟩4. □

⟨1⟩ 5 Compare the denominators.

⟨2⟩ 1 The repaired unshifted V2 comparison gives

$$\|\mathcal{U}_{N,t} - \mathcal{U}_t\|_2 \leq C(1+t)(1+L)\varepsilon_N \mathcal{L}_N^{3/2}. \quad (3.3)$$

Because: Its fourth-chaos computation contains the one-edge and two-edge classes and the nonzero four-edge class. The last contributes the explicit positive term

$$\Delta_{4e} = 4!(2Lt)^{-2} \frac{3}{2} \sum_{k_{0,1}+\dots+k_{0,4}=0} \prod_{i=1}^4 c_\times(k_{0,i}, \pi/\varepsilon_N), \quad (3.4)$$

bounded by $9t \coth(2t_0)^3 \varepsilon_N^5 / (256L^2)$. Hence it lies below the common rate in (3.3); it may not be deleted from the exact identity.

⟨2⟩ 2 The plain lattice and continuum Nelson bounds and $|e^{-x} - e^{-y}| \leq |x - y|(e^{-x} + e^{-y})$ imply

$$|\mathcal{E}_N - \mathcal{E}_\infty| \leq C(1+t)(1+L)\varepsilon_N \mathcal{L}_N^{3/2}. \quad (3.5)$$

Because: Apply Cauchy–Schwarz to the product of the exponent difference and $e^{-\mathcal{U}_N} + e^{-\mathcal{U}}$; (3.3) controls the first factor and the uniform L^2 Nelson bound the second.

⟨2⟩ 3 This proves ⟨1⟩5. □

⟨1⟩ 6 Compare the numerators.

⟨2⟩ 1 Split

$$\begin{aligned} |\mathcal{N}_N - \mathcal{N}_\infty| &\leq \mathbb{E}[|e^{i\Phi_N} - e^{i\Phi}| e^{-\mathcal{U}_{N,t}(\eta_N + h^{(N)})}] \\ &\quad + \mathbb{E}[|e^{-\mathcal{U}_{N,t}(\eta_N + h^{(N)})} - e^{-\mathcal{U}_t(\eta + h^{(\infty)})}|]. \end{aligned} \quad (3.6)$$

⟨2⟩ 2 The first term in (3.6) is at most $C_\xi 2^{-r\eta_D/4}$.

Because: Apply Cauchy–Schwarz, (2.11), and the uniform shifted lattice L^2 Nelson bound.

⟨2⟩ 3 The repaired shifted V2 theorem gives

$$\|\mathcal{U}_{N,t}(\eta_N + h^{(N)}) - \mathcal{U}_t(\eta + h^{(\infty)})\|_2 \leq C_\xi \left(2^{-r(1+\eta_D)/4} + \varepsilon_N \mathcal{L}_N^{3/2} \right). \quad (3.7)$$

Because: Decompose it into the unshifted difference (3.3) and the difference of the two shift increments. The latter is proved in the repaired V2 module by: (i) folding the lattice power symbols into the fundamental band; (ii) defining the exact-output cross and continuum symbols; (iii) isolating the mixed alias term $A_{\times,j}$; (iv) bounding filter, dispersion, high-output, and physical edge remainders separately; and (v) applying the convolution telescope to every deterministic coefficient power. Thus no undefined R_{tail} is used in (3.7).

⟨2⟩ 4 The second term in (3.6) is bounded by the right side of (3.7).

Because: Apply the exponential-difference inequality, Cauchy–Schwarz, (3.7), and the paired uniform shifted Nelson bounds: V2 for finite N , V3 for the continuum member.

⟨2⟩ 5 Therefore

$$|\mathcal{N}_N - \mathcal{N}_\infty| \leq C_\xi (2^{-r\eta_D/4} + \varepsilon_N \mathcal{L}_N^{3/2}), \quad (3.8)$$

and

$$|\mathcal{N}_N|, |\mathcal{N}_\infty| \leq B_1 < \infty. \quad (3.9)$$

Because: The slower of the two powers of 2^{-r} is $\eta_D/4$. Equation (3.9) follows by deleting the unit-modulus phase and applying the L^1 shifted Nelson bounds.

⟨2⟩ 6 This proves ⟨1⟩6. □

⟨1⟩ 7 Prove fixed-time convergence of the interacting thermal functionals.

⟨2⟩ 1 The exact algebraic difference is

$$\begin{aligned} F_N - F_\infty &= (e^{-S_N} - e^{-S_\infty}) \frac{\mathcal{N}_N}{\mathcal{E}_N} + e^{-S_\infty} \frac{\mathcal{N}_N - \mathcal{N}_\infty}{\mathcal{E}_N} \\ &\quad + e^{-S_\infty} \mathcal{N}_\infty \frac{\mathcal{E}_\infty - \mathcal{E}_N}{\mathcal{E}_N \mathcal{E}_\infty}. \end{aligned} \quad (3.10)$$

Because: Put (1.6) over the common denominators and add/subtract the two intermediate terms shown.

⟨2⟩ 2 Equations (2.7), (3.2), (3.5), (3.8), and (3.9) give

$$|F_N(t; \xi) - F_\infty(t; \xi)| \leq C_\xi (2^{-r\eta_D/4} + \varepsilon_N \mathcal{L}_N^{3/2}) \rightarrow 0. \quad (3.11)$$

⟨2⟩ 3 For every fixed $t \geq 1/L$, (3.11) is admissible by taking $t_0 = \max(t/2, 1/L)$ and $T = 2t$.

Because: Then $t_0 \leq t \leq T$ and $t_0 L \geq 1$. This is the genuine range of the V2 comparison; no conclusion at arbitrarily small t is inferred.

⟨2⟩ 4 This proves ⟨1⟩7 (A6). □

5 Partition functions and exact Laplace transfer

⟨1⟩ 8 Compare the normal-ordered free partition functions.

⟨2⟩ 1 For every $t > 0$,

$$\tilde{Z}_N^{(0)}(t) := \text{Tr} e^{-t(H_{N,L}^{(0)} - e_0(N))} = \prod_{k \in \Gamma_N} (1 - e^{-t\gamma_N(k)})^{-1}. \quad (4.1)$$

Because: Subtracting $e_0(N) = \frac{1}{2} \sum_k \gamma_N(k)$ removes the zero-point energy from the product of one-oscillator traces.

⟨2⟩ 2 For $k \in \Gamma_N$, the mean-value theorem gives

$$|\log(1 - e^{-t\gamma_N(k)}) - \log(1 - e^{-t\gamma(k)})| \leq \frac{te^{-t\gamma_N(k)}}{1 - e^{-tm}} |\gamma(k) - \gamma_N(k)|. \quad (4.2)$$

Because: The derivative of $y \mapsto \log(1 - e^{-ty})$ is $t/(e^{ty} - 1) \leq te^{-ty}/(1 - e^{-tm})$ for $y \geq m$, and $\gamma_N \leq \gamma$.

⟨2⟩ 3 Using

$$0 \leq \gamma(k) - \gamma_N(k) \leq \frac{\varepsilon_N^2 k^4}{24m}, \quad \gamma_N(k) \geq \max(m, 2|k|/\pi), \quad (4.3)$$

the sum of (4.2) is $O(\varepsilon_N^2)$.

Because: The remaining series $\sum_k e^{-t \max(m, 2|k|/\pi)} k^4$ converges with a bound independent of N .

⟨2⟩ 4 The omitted continuum modes contribute $O(e^{-2t/\varepsilon_N})$.

Because: The asymmetric complement begins at magnitude π/ε_N , including the positive edge. Use $|\log(1 - e^{-x})| \leq e^{-x}/(1 - e^{-tm})$ and $\gamma(k) \geq |k|$; sum the geometric spatial tail.

⟨2⟩ 5 Hence

$$|\log \tilde{Z}_N^{(0)}(t) - \log Z_0(t)| \leq C(m, L, t) \varepsilon_N^2, \quad \tilde{Z}_N^{(0)}(t) \rightarrow Z_0(t). \quad (4.4)$$

⟨2⟩ 6 This proves ⟨1⟩ 8. □

⟨1⟩ 9 Compare the interacting normal-ordered partition functions and obtain a uniform lower ground-energy bound.

Fix $t_* > 0$ with $t_* L \geq 1$.

⟨2⟩ 1 For every $t \geq t_*$,

$$\tilde{Z}_N(t) := \text{Tr} e^{-t(K_{N,L} - e_0(N))} = \tilde{Z}_N^{(0)}(t) \mathcal{E}_N(t). \quad (4.5)$$

Because: Apply the finite-dimensional cyclic Mehler trace construction from V1/V3 to the lattice dispersions and interaction. Normal ordering removes only the free zero-point factor. The repaired V2 proposition identifies the cyclic Gaussian law, including the symmetrized edge column, and Wick recombination identifies the path integrand with $\mathcal{U}_{N,t}$.

⟨2⟩ 2 Equations (3.5), (4.4), and the V3 continuum partition identity imply

$$\tilde{Z}_N(t) \rightarrow Z_0(t) \mathcal{E}_\infty(t) = \text{Tr} e^{-tK_L} =: Z_\infty(t) \quad (t \geq t_*). \quad (4.6)$$

Because: For this fixed t , use the admissible interval $t_0 = \max(t/2, 1/L)$, $T = 2t$ in (3.5). This avoids the invalid choice $t_0 = t/2$ when $t < 2/L$.

⟨2⟩ 3 There is $c_\lambda < \infty$ such that

$$\inf_N \tilde{E}_0(N) \geq -c_\lambda. \quad (4.7)$$

Because: Fix any $t_1 \geq t_*$. Then

$$e^{-t_1 \tilde{E}_0(N)} \leq \tilde{Z}_N(t_1) = \tilde{Z}_N^{(0)}(t_1) \mathcal{E}_N(t_1).$$

The first factor is uniformly bounded by (4.4), and the second by the unshifted L^1 Nelson bound. Take logarithms and divide by t_1 .

⟨2⟩ 4 This proves ⟨1⟩9. □

⟨1⟩ 10 Prove the Laplace-transfer lemma in the exact range needed here.

Let $\nu_N = \sum_{j \geq 0} \delta_{x_j(N)}$ be locally finite eigenvalue point measures, assume all $x_j(N) \geq c$, and suppose

$$\int e^{-tx} d\nu_N(x) \rightarrow \int e^{-tx} d\nu_\infty(x) \quad \text{for every } t \geq t_*. \quad (4.8)$$

⟨2⟩ 1 The span $\mathcal{A}_{t_*}$ of $x \mapsto e^{-tx}$, $t \geq t_*$, is uniformly dense in $C_0([c, \infty))$.

Because: It is a self-adjoint algebra: products add the two parameters and remain at least t_* . It separates points and vanishes nowhere. The locally compact Stone–Weierstrass theorem applies.

⟨2⟩ 2 For $f \in C_c([c, \infty))$, put $\tilde{f}(x) = f(x)e^{t_*x}$ and choose $g \in \mathcal{A}_{t_*}$ with $\|\tilde{f} - g\|_\infty < \delta$. Then

$$\left| \int f d\nu_N - \int g e^{-t_*x} d\nu_N \right| \leq \delta \int e^{-t_*x} d\nu_N. \quad (4.9)$$

Because: Multiply the pointwise approximation error by the finite measure $e^{-t_*x} d\nu_N$. This weighted step is necessary: ν_N itself generally has infinite total mass.

⟨2⟩ 3 The right side of (4.9) is uniformly $O(\delta)$, and the middle integrals converge.

Because: Equation (4.8) at t_* makes their masses uniformly bounded. Since g is a finite linear combination of e^{-tx} with $t \geq t_*$, ge^{-t_*x} is a finite linear combination with parameters at least $2t_*$, so (4.8) applies.

⟨2⟩ 4 Let $N \rightarrow \infty$ and then $\delta \downarrow 0$ in (4.9). Thus $\nu_N \rightarrow \nu_\infty$ vaguely.

⟨2⟩ 5 Vague convergence implies convergence of every ordered atom with multiplicity:

$$x_j(N) \rightarrow x_j(\infty). \quad (4.10)$$

Because: Around the first $j+1$ limit atoms choose finitely many pairwise disjoint intervals whose endpoints are not atoms and which contain no other limit atoms. Continuous upper and lower approximations to their indicators show that the integer masses in each interval eventually equal the limiting multiplicities. Shrink the intervals. Induction in j yields (4.10).

⟨2⟩ 6 For $t \geq t_*$,

$$D_N(t) = e^{tx_0(N)} \int e^{-tx} d\nu_N(x) - 1 \rightarrow D_\infty(t). \quad (4.11)$$

Because: Combine (4.8) and (4.10) for $j = 0$. No conclusion is asserted for $t < t_*$; the hypotheses supply no control there.

⟨2⟩ 7 This proves ⟨1⟩10. □

⟨1⟩ 11 Apply the transfer lemma to the interacting spectra.

⟨2⟩ 1 Take $x_j(N) = \tilde{E}_j(N)$ and $x_j(\infty) = \mu_j$. Equations (4.6) and (4.7) verify (4.8) and the common lower support bound.

⟨2⟩ 2 Therefore, for every fixed j ,

$$\tilde{E}_j(N) \rightarrow \mu_j. \quad (4.12)$$

⟨2⟩ 3 In particular,

$$E_1(N) - E_0(N) - \mu_1 - \mu_0 = \gamma_L > 0, \quad (4.13)$$

and for every $t \geq t_*$,

$$D_N(t) \rightarrow D_\infty(t) < \infty. \quad (4.14)$$

Because: The zero-point shift cancels from each eigenvalue difference and from D_N . Apply (4.10)–(4.11) and the V3 strict gap.

⟨2⟩ 4 This proves ⟨1⟩11 (A7). □

6 Thermal-to-ground-state interchange and the gate

⟨1⟩ 12 Prove the spectral tail estimate.

Let K have discrete spectrum $E_0 < E_1 \leq \dots$, normalized ground vector Ω , and let W be unitary.

⟨2⟩ 1 With $Z = e^{-tE_0}(1 + D(t))$,

$$\frac{\text{Tr}(e^{-tK}W)}{Z} - \langle \Omega, W\Omega \rangle = \frac{1}{Z} \sum_{j \geq 1} e^{-tE_j} (\langle \phi_j, W\phi_j \rangle - \langle \Omega, W\Omega \rangle). \quad (5.1)$$

Because: Expand the trace in an orthonormal eigenbasis and cancel the $j = 0$ term.

⟨2⟩ 2 Since every displayed matrix element has modulus at most one,

$$\left| \frac{\text{Tr}(e^{-tK}W)}{\text{Tr } e^{-tK}} - \langle \Omega, W\Omega \rangle \right| \leq \frac{2D(t)}{1 + D(t)} \leq 2D(t). \quad (5.2)$$

⟨2⟩ 3 This applies to every lattice $K_{N,L}$ and to K_L .

Because: Lattice compactness and ground simplicity were proved in the model sheet; V3 proves continuum trace class, discreteness, and ground simplicity.

⟨2⟩ 4 This proves ⟨1⟩12. □

⟨1⟩ 13 Prove the abstract interchange of fixed-temperature and ground-state limits.

Suppose $a_N(t)$, ℓ_N , and $D_N(t)$ obey

$$|a_N(t) - \ell_N| \leq c_0 D_N(t), \quad |a_N(t)| \leq c_0, \quad (5.3)$$

$a_N(t) \rightarrow a_\infty(t)$ for each $t \geq t_0$, $D_N(t_1) \rightarrow D_\infty(t_1) < \infty$ at one $t_1 \geq t_0$, and

$$\liminf_N (E_1(N) - E_0(N)) \geq \gamma > 0. \quad (5.4)$$

⟨2⟩ 1 For $t \geq t_1$,

$$D_N(t) \leq e^{-(t-t_1)(E_1(N)-E_0(N))} D_N(t_1). \quad (5.5)$$

Because: Factor $e^{-(t-t_1)(E_j-E_0)}$ from every summand and bound it by the $j = 1$ value.

⟨2⟩ 2 There are $N_0, D_* < \infty$ such that, for all $N \geq N_0$,

$$D_N(t) \leq D_* e^{-(t-t_1)\gamma/2}. \quad (5.6)$$

Because: Convergence at t_1 bounds $D_N(t_1)$; the liminf in (5.4) gives an eventual gap at least $\gamma/2$.

⟨2⟩ 3 The continuum tail independently satisfies

$$D_\infty(t) \leq D_\infty(t_1) e^{-(t-t_1)\gamma_L} \rightarrow 0. \quad (5.7)$$

⟨2⟩ 4 Given $\delta > 0$, choose t so that the right sides of (5.6)–(5.7) are below δ . Then

$$|\ell_N - \ell_\infty| \leq 2c_0\delta + |a_N(t) - a_\infty(t)|. \quad (5.8)$$

Because: Add and subtract $a_N(t), a_\infty(t)$ and apply (5.3) to the two tails.

⟨2⟩ 5 Let $N \rightarrow \infty$ in (5.8), then $\delta \downarrow 0$. Thus $\ell_N \rightarrow \ell_\infty$.

⟨2⟩ 6 This proves ⟨1⟩13. □

⟨1⟩ 14 Close the Weyl gate at every fixed coarse scale.

Set

$$t_* = \max(1, 1/L), \quad a_N(t) = F_N(t; \xi), \quad \ell_N = \omega_N(W_N(R_M^N \xi)), \quad (5.9)$$

with the corresponding continuum definitions.

⟨2⟩ 1 The bound (5.3) holds with $c_0 = 2$.

Because: $|F_N| \leq 1$ and the spectral estimate (5.2) applies on both sides.

⟨2⟩ 2 For each fixed $t \geq t_*$, $a_N(t) \rightarrow a_\infty(t)$.

Because: $t_* \geq 1/L$, so Step ⟨1⟩7 applies.

⟨2⟩ 3 $D_N(t_*) \rightarrow D_\infty(t_*) < \infty$.

Because: Step ⟨1⟩11 applies with the admissible lower time t_* .

⟨2⟩ 4 $\liminf_N (E_1(N) - E_0(N)) = \gamma_L > 0$.

Because: This is (4.13).

⟨2⟩ 5 Step ⟨1⟩13 therefore gives

$$\lim_{N \rightarrow \infty} \omega_N(\alpha_M^N(W_M(\xi))) = \omega_{L,\lambda}^{\text{ct}}(\beta_M(W_M(\xi))). \quad (5.10)$$

Because: The embeddings are defined by $\alpha_M^N(W_M(\xi)) = W_N(R_M^N \xi)$ and $\beta_M(W_M(\xi)) = W_{\text{ct}}(R_M^\infty \xi)$.

⟨2⟩ 6 This proves ⟨1⟩14 (the gate). □

7 Inductive-limit state, identification, and symmetry

⟨1⟩ 15 Construct the continuum embedding of the inductive-limit algebra.

Assume the already proved scaling identity $\beta_N \alpha_M^N = \beta_M$ and injectivity of all maps.

⟨2⟩ 1 On the algebraic union $\bigcup_M \alpha_M^\infty(\mathfrak{W}_M)$ define

$$\beta(\alpha_M^\infty(A)) = \beta_M(A). \quad (6.1)$$

⟨2⟩ 2 Definition (6.1) is independent of the representative.

Because: If $M \leq M'$ and $\alpha_M^\infty(A) = \alpha_{M'}^\infty(A')$, injectivity of $\alpha_{M'}^\infty$ gives $A' = \alpha_M^{M'}(A)$. Hence $\beta_{M'}(A') = \beta_{M'} \alpha_M^{M'}(A) = \beta_M(A)$.

⟨2⟩ 3 β is isometric on the union.

Because: An injective $*$ -homomorphism between C^* -algebras is isometric; thus $\|\beta(\alpha_M^\infty(A))\| = \|\beta_M(A)\| = \|A\| = \|\alpha_M^\infty(A)\|$.

⟨2⟩ 4 It extends uniquely to an injective $*$ -homomorphism

$$\beta : \mathfrak{W}_\infty \longrightarrow \mathfrak{W}_{\text{ct}}, \quad \beta \alpha_M^\infty = \beta_M. \quad (6.2)$$

Because: The algebraic union is dense in its inductive-limit completion. An isometry extends uniquely; multiplication and involution pass by continuity, and an isometry is injective.

⟨2⟩ 5 This proves ⟨1⟩15. □

⟨1⟩ 16 Apply the previously proved Weyl extension criterion.

Define

$$\chi_M(\xi) = \lim_{N \rightarrow \infty} \omega_N(\alpha_M^N(W_M(\xi))). \quad (6.3)$$

⟨2⟩ 1 The limit exists and equals

$$\chi_M(\xi) = \omega_{L,\lambda}^{\text{ct}}(\beta_M(W_M(\xi))). \quad (6.4)$$

Because: This is (5.10).

⟨2⟩ 2 For each M , there is a unique state ρ_M on $\mathfrak{W}_M$ satisfying $\rho_M(W_M(\xi)) = \chi_M(\xi)$, and

$$\omega_N(\alpha_M^N(A)) \rightarrow \rho_M(A) \quad (A \in \mathfrak{W}_M). \quad (6.5)$$

Because: The Weyl-reduction module proves: pointwise convergence on all Weyl generators, uniform state norm one, and density of finite Weyl linear combinations extend the limit uniquely to the whole C^* -algebra. Equation (6.4) also supplies positive definiteness and continuity directly as a restriction of a state.

⟨2⟩ 3 The family is consistent:

$$\rho_N \circ \alpha_M^N = \rho_M. \quad (6.6)$$

Because: Check it on $W_M(\xi)$ using $\alpha_M^N(W_M(\xi)) = W_N(R_M^N \xi)$ and the compatibility of the continuum refinement; extend by norm density.

⟨2⟩ 4 There is a unique state Ω_∞ on $\mathfrak{W}_\infty$ with

$$\Omega_\infty \circ \alpha_M^\infty = \rho_M, \quad \Omega_\infty = \omega_{L,\lambda}^{\text{ct}} \circ \beta. \quad (6.7)$$

Because: Consistency defines a norm-one positive functional on the dense algebraic union, hence a unique state on the completion. Equations (6.2), (6.4) show that it agrees there with the continuum pullback; density gives the second identity.

⟨2⟩ 5 All states in (6.5)–(6.7) are invariant under the field reflection $\vartheta(W(\zeta)) = W(-\zeta)$.

Because: Every refinement map is linear, hence intertwines ϑ . The lattice Hamiltonian is even and commutes with its reflection unitary. Ground-state simplicity forces that unitary to multiply Ω_N by a phase, so $\omega_N \circ \vartheta_N = \omega_N$. Pass to the limits in (6.5), then use consistency. Equivalently, the earlier Weyl-reduction symmetry proposition gives the same conclusion.

⟨2⟩ 6 This proves ⟨1⟩16 and the full fixed-coarse theorem. □

8 V4 conclusion and validity status

Step ⟨1⟩7 proves fixed- t thermal convergence with rate. Steps ⟨1⟩8–⟨1⟩11 prove normal-ordered partition convergence, eigenvalue convergence, the eventual lattice gap, and tail convergence. Steps ⟨1⟩12–⟨1⟩14 interchange the thermal and ground limits and prove the gate. Steps ⟨1⟩15–⟨1⟩16 identify the unique projective limit with the continuum vacuum and carry the $\mathbb{Z}_2$ symmetry.

This conclusion is conditional on replacing the original V2 Lemma 6.4/Proposition 6.5 proof by the exact repaired comparison in the preceding Lamport module. The original Part II files, without that replacement and the four-edge correction, do not prove their final theorem as written.
